

Storage

Store below 30°C. Protect from light. Do not freeze. Do not use if the solution is discoloured.  
Once the foil pouch is opened the respules should be used within three month and should be stored in temperature not exceeding 25°C.

Shelf life

24 Months

Packing/Pack Size

Packing : Each pack contains 20 respules of 2.5 ml each.  
Pack size: Carton containing pouch having respule of 2.5 ml

Registration Holder in Malaysia

CIPLA MALAYSIA SDN BHD  
Suite 1101, Amcorp Tower, Amcorp Trade Centre,  
18 Persiaran Barat, 46 050 Petaling Jaya, Selangor, Malaysia.

Mfd. by CIPLA LTD.

Plot No. L-139 to L-146,  
Verna Industrial Estate,  
Verna- Goa INDIA

Date of Revision: March 2017

**Directions for use:** The product should be used with a respirator or nebuliser only, under the direction of a physician. It is notto be injected or administered orally.

Remove a pouch of combipack respule from the carton.



Tear one respule from the combipack



To open the respule twist off the cap as shown in the figure.



Squeeze the content into the reservoir of the nebulizing chamber.



Discard the respule after use.

Use a fresh respule for each dose.

Store the respules in foil pouch at all times.

For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory

Ipratropium Bromide and Salbutamol Respirator solution

Duolin Respules

Composition  
Each 2.5 ml respule contains  
Ipratropium Bromide BP  
equivalent to Ipratropium Bromide  
(anhydrous)..... 500mcg  
Salbutamol Sulphate BP  
equivalent to Salbutamol..... 2.5 mg  
in an Isotonic solution..... q.s.

Description

A clear, colourless solution filled in 2.5 ml FFS respule. On visual examination, there is no sign of leakage or physical damage.

Pharmacodynamics

Duolin contains two active bronchodilating substances, salbutamol sulphate and ipratropium bromide.

Salbutamol sulphate is a beta<sub>2</sub>-adrenergic agent which acts on airway smooth muscle resulting in relaxation. Salbutamol relaxes all smooth muscle from the trachea to the terminal bronchioles and protects against all bronchoc-onstrictor challenges.

Ipratropium bromide is a quaternary ammonium compound with anticholin-ergic properties. In preclinical studies, it appears to inhibit vagally mediated reflexes by antagonising the action of acetylcholine, the transmitter agent released from the vagus nerve. Anticholinergics prevent the increase of intracellular concentration of cyclic guanosine monophosphate (cyclic GMP) caused by interaction of acetylcholine with muscarinic receptors on bron-chial smooth muscle. The bronchodilation following inhalation of ipratropium bromide is primarily local and site specific to the lung and not systemic in nature.

Duolin provides the simultaneous release of ipratropium bromide and salbutamol allowing the synergistic efficacy on the muscarinic and beta<sub>2</sub>-adrenergic receptors in the airways to cause bronchodilation which is su-perior to that provided by each single agent and with no potentiation of adverse events.

Pharmacokinetics

Ipratropium bromide is quickly absorbed after oral inhalation.The systemic bioavailability following inhalation is estimated to be less than 10% of the dose. Renal excretion of ipratropium bromide is given as 46% of the dose after intravenous administration. The half-life of the terminal elimination phase is about 1.6 hours as determined after intravenous administration. The half-life elimination of drug and metabolites is 3.6 hours, as deter-mined after radio labelling. Ipratropium bromide does not penetrate the blood brain barrier.

Salbutamol sulphate is rapidly and completely absorbed following adminis-tration either by the inhaled or oral route. Peak plasma salbutamol concen-trations are seen within three hours of administration and it is excreted unchanged in the urine after 24 hours. The elimination half-life is 4 hours.Salbutamol will cross the blood brain barrier reaching concentrations amounting to about five per cent of the plasma concentrations.

It has been shown that co-nebulisation of ipratropium bromide and salbutamol sulphate does not potentiate the systemic absorption of either component and that therefore the additive activity of Duolin is due to the combined local effect on the lung following inhalation.

Indication

Duolin is indicated for the treatment of reversible bronchospasm associated with obstructive airway diseases in patients who require more than a single bron-chodilator.

Dosage

The dosage should be adapted to the individual requirements of the patient.

1 Respule nebulised three or four times daily.

Children - Not recommended

Contraindications

Duolin is contraindicated in patients with hypertrophic obstructive cardi-omyopathy and tachyarrhythmia and in patients with a history of hypersen-sitivity to atropine or its derivatives, or to any other component of the prod-uct.

Warning & Precautions

In the case of acute, rapidly worsening dyspnoea a doctor should be con-sulted immediately.

Immediate hypersensitivity reactions may occur after administration of Duolin as demonstrated by rare cases of urticaria, angioedema, rash, broncho-spasm and oropharyngeal oedema.

Ocular complications

There have been isolated reports of ocular effects, e.g. mydriasis, blurring of vision, eye pain, increased intraocular pressure, narrow-angle glaucoma, eye pain when aerosolised ipratropium bromide either alone or in combina-tion with an adrenergic beta<sub>2</sub>-agonist containing ipratropium bromide have escaped into the eyes.

Eye pain or discomfort, blurred vision, visual halos or coloured images in association with red eyes from conjunctival congestion and corneal oedema may be signs of acute narrow-angle glaucoma. Should any combination of these symptoms develop, treatment with miotic drops should be initiated and specialist advice sought immediately.

In the following situations Duolin should only be used after careful risk / benefit assessment, especially when doses higher than recommended are used:

- insufficiently controlled diabetes mellitus
- recent myocardial infarction
- severe organic heart or vascular disorders
- hyperthyroidism
- phaeochromocytoma
- risk of narrow-angle glaucoma
- prostatic hypertrophy
- bladder-neck obstruction

Potentially serious hypokalaemia may result from prolonged and / or high dose beta<sub>2</sub>-agonist therapy. Additionally, hypoxia may aggravate the ef-fects of hypokalaemia on cardiac rhythm. It is recommended that serum potassium levels be monitored in such situations.

Patients with cystic fibrosis may be more prone to gastrointestinal motility disturbances.

If higher than recommended doses of Duolin are required to control symp-toms, the patient's therapy plan should be reviewed by a doctor.

Interaction with other Medicaments

The concurrent administration of other beta-mimetics, systemically absorbed anticholinergics and xanthine derivatives may increase the side effects.

Beta-agonist induced hypokalaemia may be increased by concomitant treat-ment with xanthine derivatives, glucocorticosteroids and diuretics. This should be taken into account particularly in patients with severe airway obstruction.

Hypokalaemia may result in an increased susceptibility to arrhythmias in pa-tients receiving digoxin. It is recommended that serum potassium levels be monitored in such situations.

A potentially serious reduction in bronchodilator effect may occur during concur-rent administration of beta-blockers.

Beta-adrenergic agonists should be administered with caution to patients being treated with monoamine oxidase inhibitors or tricyclic antidepressants, since the action of beta-adrenergic agonists may be enhanced.

Inhalation of halogenated hydrocarbon anaesthetics such as halothane,trichloroethylene and enflurane may increase the susceptibility to the cardiovascular effects of beta-agonists.

Pregnancy and Lactation

Although there is little published evidence of the safety of Duolin in the early stages of pregnancy, the individual active ingredients, ipratropium bromide and salbutamol have been in widespread use for many years. No suspect teratogenic factors have been observed in general practice or clinical studies. As a general precaution, however, Duolin should only be used in pregnancy and during the lactation period if the expected benefit to the mother is greater than any possible risk to the child.

Duolin should be used with caution before childbirth in view of salbutamol's inhibitory effects on uterine contractions.

Salbutamol sulphate and ipratropium bromide are probably excreted in breast milk and their effects on neonates are not known. Although lipid-insoluble quaternary bases pass into breast milk, it is unlikely that this will happen to any extent especially when taken by inhalation. However, be-cause many drugs are excreted in breast milk, caution should be exercised when Duolin is administered to a nursing woman.

Side Effects

In common with other beta-agonists containing products, side effects of Duolin can include fine tremor of skeletal muscles and nervousness and less frequently, tachycardia, dizziness, palpitations or headache, especially in hypersensitive patients.

Potentially serious hypokalaemia may result from prolonged and / or high dose beta<sub>2</sub>-agonist therapy.

As with other beta-mimetics, nausea, vomiting, sweating, weakness and myalgia/muscle cramps may occur. In rare cases decrease in diastolic blood pressure, increase in systolic blood pressure, arrhythmias, particularly af-ter higher doses, may occur.

In individual cases psychological alterations have been reported under in-halation therapy with beta-mimetics.

The most frequent non-respiratory anticholinergic related adverse events are dryness of mouth and dysphonia.

There have been isolated reports of ocular complications (i.e. mydriasis, increased intraocular pressure, angle-closure glaucoma, and eye pain) when aerosolised ipratropium bromide either alone or in combination with adren-ergic beta<sub>2</sub>-agonist, has escaped into the eyes. Ocular side effects, gas-trointestinal motility disturbances and urinary retention may occur in rare cases and are reversible.

Symptoms and Treatment of Overdose

The effects of overdosage are expected to be primarily related to salbutamol because acute overdosage with ipratropium bromide is unlikely as it is not well absorbed systemically after inhalation or oral administration.

Symptoms

Manifestations of overdosage with salbutamol may include tachycardia, anginal pain, hypertension, hypotension,palpitations, tremor, widening of the pulse pres-sure, arrhythmia and flushing.

Therapy

Administration of sedatives, tranquillisers.

In severe cases, intensive therapy

Beta-receptor blockers, preferably beta<sub>2</sub>-selective, are suitable as specific antidotes; however, a possible increase in bronchial obstruction must be taken into account and the dose should be adjusted carefully in patients suffering from bronchial asthma.

Cipla

PACKAGING DEVELOPMENT

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| Product Name: Duolin Respules                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |                                | Material No.: xxxxxxxx | Version: 01                 | Item: Leaflet            | Co-ordinator: Vinod                  | Artist: Vasundhara    | Date: 23-05-2017                                                                                     |
| <div>Colours:<br/>BLUE WOOL TEST VALUE 5-8<br/>(LIGHT FASTENING DATA)</div> <div><div></div>Black</div>                                                                                                                                                                                                                                                                                                                                                                                    |  |                                |                        |                             |                          | INK: Oil based Ink from DIC OR MICRO |                       |                                                                                                      |
| Design: Folded                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |                                |                        | Supersedes/Reference: JE24K |                          | Software: Illustrator CC             |                       |                                                                                                      |
| Fonts: -----                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |                                |                        |                             | Links: xxxxxx            |                                      |                       |                                                                                                      |
| Actual Size: 170 x 180 mm                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  | Size after folding: 85 x 30 mm |                        | PharmaCode: 1121 Mini.      |                          | Grain Direction : Parallel to length |                       | Screen : # ____                                                                                      |
| Material: 54 GSM Maplitho Paper                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |                                |                        |                             |                          | Varnish:                             |                       | Artwork Print Size:<br><div><input type="checkbox"/> actual    <input type="checkbox"/> scaled</div> |
| Path: PC:/D:/Vasundhara/Malaysia/Duolin/xxxxxxx Duolin Respules Leaflet Malaysia.ai                                                                                                                                                                                                                                                                                                                                                                                                        |  |                                |                        |                             |                          |                                      |                       |                                                                                                      |
| <div>• <b>Instructions / Remark: ----</b></div> <div>• Any deviation must be brought to the notice of packaging development co-ordinator immediately.</div> <div>• For any clarification, please contact packaging development co-ordinator immediately.</div> <div>NO CHANGES IN ARTWORK SHOULD BE DONE BY THE PRINTER</div> <div>• The printer should verify the e-proof against the approved artwork before submitting for approval and the e-proof should have printer details .</div> |  |                                |                        | Checked by                  | Artist                   | Cordinator                           | file loaded in Server | Section Head                                                                                         |
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